

UNIT-1 SEMICONDUCTOR DEVICES

20-08-26

~ Semiconductors are materials whose electrical properties lie between conductors and insulators.

Ex: Silicon and Germanium

~ Silicon and Germanium are the frequently used for semiconducting materials.

~ These Semiconductors are mainly used for electronic components and photo-voltaic cells.

~ Properties of Semiconductors

- i) A semiconductor is formed by covalent bond.
- ii) A Semiconductor has crystalline structure
- iii) With the presence of impurity resistivity of this material decreases very drastically.
- iv) Semiconductor is tetravalent ($4e^-$ in outermost shell)
- v) Semiconductor has resistivity value that lies between the resistivity of the insulator and conductor (10^6 to $0.5 \Omega \cdot m$)

vi) Forbidden energy gap for a semiconductor is very less (0.7 to $1.1 eV$)

vii) Semiconductor has negative temperature coefficient.

~ Types of Semiconductors

There are two types of semiconductors

- i) Intrinsic Semiconductors
- ii) Extrinsic Semiconductors

→ Intrinsic Semiconductors:

A semiconductor, which is in its extremely pure form is known as Intrinsic Semiconductor.

→ Extrinsic Semiconductors:

- Semiconductors which are formed after adding impurities are called extrinsic Semiconductors.

- Intrinsic + impurity = extrinsic

- The addition of impurity to semiconductor is called doping.

- Sometimes, impurity is called as a dopant.

- Two types of impurities may be added to intrinsic Semiconductors, they are

- * Donor impurity (Pentavalent impurities)

- * Acceptor impurity (Trivalent impurities)

- Donor impurity (phosphorus, antimony, arsenic and bismuth)

- Acceptor impurities (Gallium, indium, aluminium, boron)

2. Extrinsic Semiconductors are classified into two types *

- * N Type Semiconductor

- * P Type Semiconductor

* N Type Semiconductors:

when the donor impurities are added to the intrinsic semiconductor, then forms the extrinsic semiconductor. This is known as N-Type Semiconductor.

Intrinsic Semiconductor + donor impurity = N type Semi

* P Type Semiconductor:
when the acceptor impurities are added to the intrinsic Semiconductor then forms the extrinsic Semiconductor. Then this is known as P Type Semiconductor.

Intrinsic + Acceptor impurity = P type semi

- p type materials is one in which holes are majority carriers. i.e., they are positively charged materials (++++).

- N type materials is one in which electrons are majority charge carriers. i.e., they are negatively charged materials (-----).

Q- what is p-n junction and define the depletion region?

- A p-n junction is formed when a p type and n type Semiconductors are joined together.

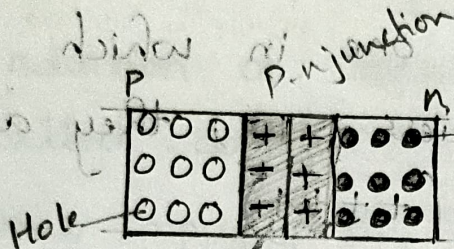
- The boundary between the p type and n type regions is called the junction

- At p-n junction, electrons from n-side move to the p side and recombine with holes.

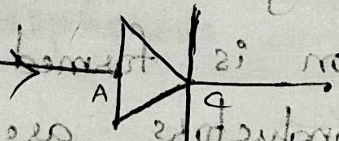
- Holes from p side similarly move to n-side, where they recombine with holes.

- As a result of this flow, the n-side has net positive charge, and p-side has net negative charge.

- Depletion region is around the junction is left, with neither holes nor free e^- .
- This neutral region which has no charge carriers is called depletion layer.
- This layer which has no charge carrier is poor conductor of electricity.

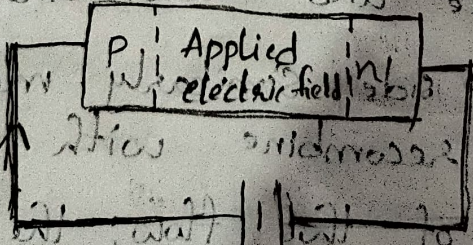


→ Diode:
 Diode is a electronic device which allows the flow of current only in one direction.

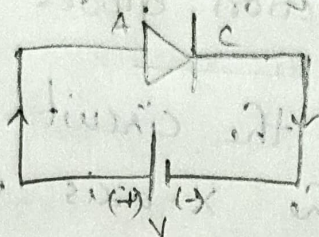


Diode contains mainly two terminals anode and cathode.

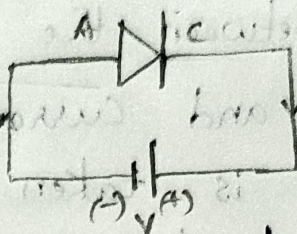
Q. Explain about the P-n junction diode operation and draw its characteristics.



- Forward Biasing



- Reverse Biasing



The p-n junction is considered biased when an external voltage is applied across it.

There are two types of biasing:

1. Forward bias

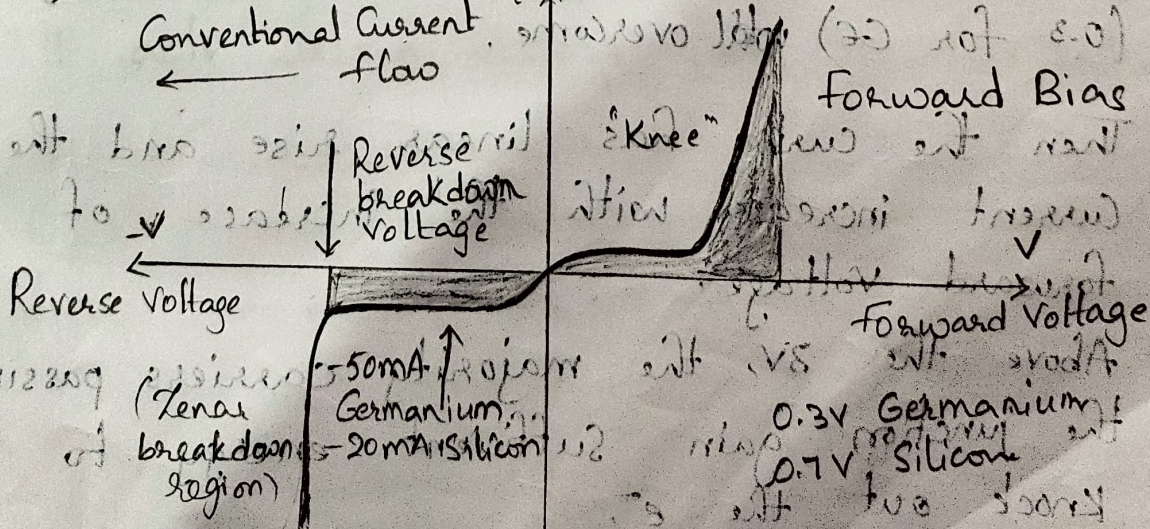
2. Reverse bias

In a p-n junction when the positive terminal is connected to the p-region and negative terminal is connected to n-region. Then that is known as Forward bias mode.

In a p-n junction when the negative terminal is connected to the p-region and positive terminal connected to n-region. Then that is known as reverse bias mode.

* Characteristics of Diodes

Conventional Current flow



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VI characteristics of P-N Junction diodes is a curve between the

- Voltage and current through the circuit
- Voltage is taken along the x-axis while the current is taken along the y-axis.
- The above graph is the V-I characteristics curve of the P-N Junction diode.

- With the help of the curve, we can understand that there are three regions in which the diode works, and they are

i) Zero bias ii) Forward bias iii) Reverse bias.

Forward biasing

When voltage is applied across a diode in such that the diode allows current and the potential barrier reduced, the diode is said to be forward biased.

In forward bias,

- No current flows until the barrier voltage V (0.3 for Ge) is overcome.
- Then the curve has linear rise and the current increases with the increase of forward voltage.
- Above the 3V, the majority carriers passing the junction gain sufficient energy to knock out the e^- .
- Therefore, the forward current increases sharply.

- Reverse biasing

when voltage is applied across a diode in such a way that the diode prohibits current and potential barrier increases, the diode is said to be reverse biased.

In Reverse bias,

- Junction resistance, potential barrier increase
- when reverse voltage is increased beyond a value, called breakdown voltage
- Reverse current increase sharply.
- Above 25V reverse voltage, destroys the junction permanently.

~ Applications of p-n junction diode :

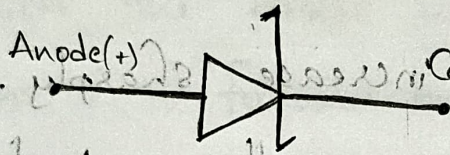
- p-n junction ^{diode} can be used as a photo diode.
- It can be used as a solar cell.
- It can be used LED lighting applications
- It can be used in rectifiers.
- It acts as a voltage controlled oscillator.

→ Zener Diode :

Zener diode is defined as the semiconductor which is heavily doped to operate in reverse direction or in breakdown region.

• Designed to operate in reverse breakdown mode.

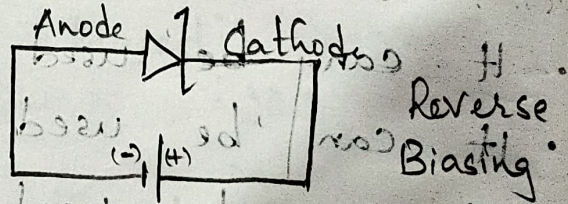
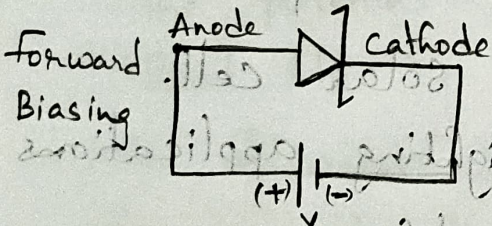
• It maintains a constant voltage across its terminals.



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Q Explain about the Zener diode operations and draw its voltage and current operations.

Zener diode circuit diagram



• Zener diode operates in two modes Forward biasing mode and reverse biasing mode.

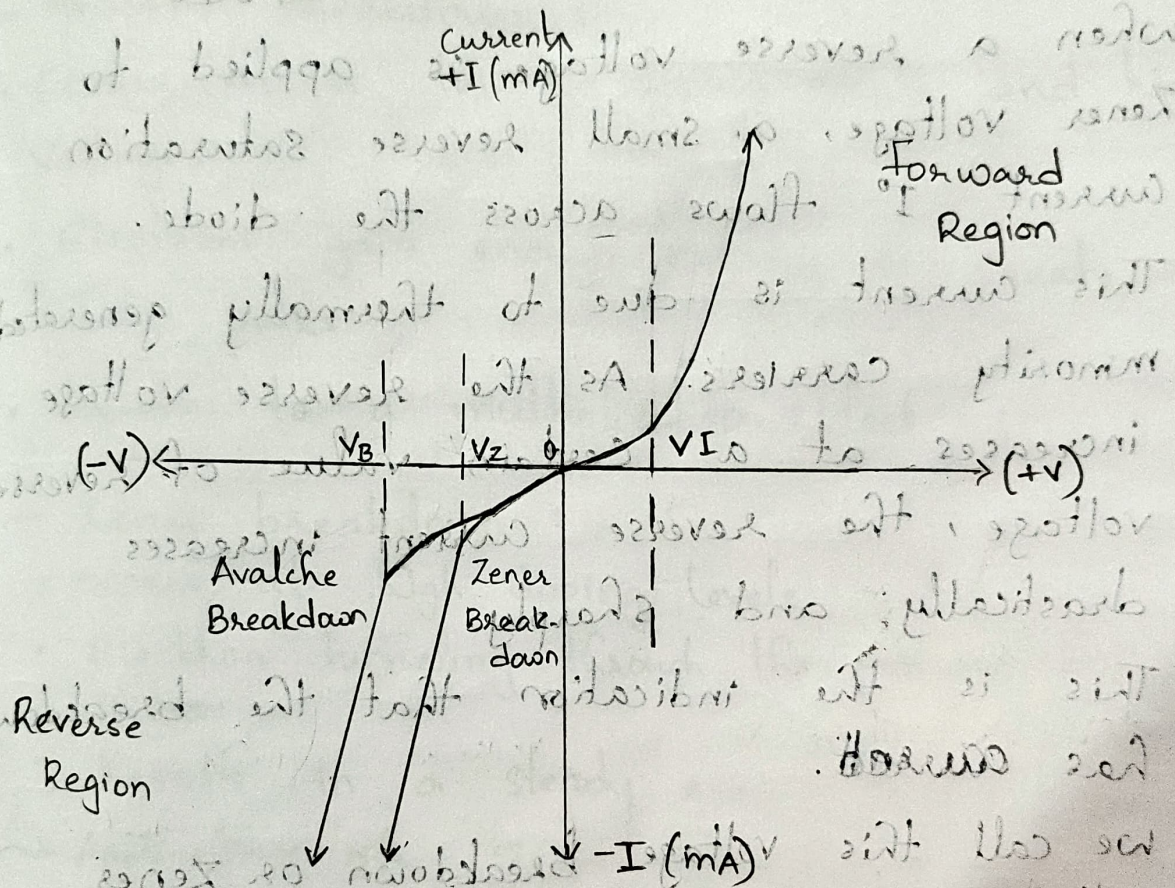
• Forward biasing : In a Zener diode, battery positive terminal is connected to anode; negative terminal is connected to cathode, this type is called as forward biasing mode.

• Reverse biasing : In a Zener diode, battery negative terminal is connected to anode, positive terminal is connected to cathode, this type is

called as reverse biasing mode.

The V-I characteristics of a Zener diode can be divided into two parts:

- i) forward characteristics
- ii) Reverse characteristics



→ Forward characteristics of Zener diode:

- The first quadrant in the graph represents the forward characteristics of Zener diode.
- No current flows until the barrier voltage (0.3 for Ge) will overcome.
- Then the curve has the linear rise and the current increase with the increase of forward voltage.

- Above the 3V, the majority carriers passing the Junction gain sufficient energy to knock out the elements.

- Therefore, the forward current increase sharply.

* Reverse characteristics of Zener diode :

- When a reverse voltage is applied to Zener voltage, a small reverse saturation current I^0 flows across the diode.

- This current is due to thermally generated minority carriers. As the reverse voltage

- increases at a certain value of reverse voltage, the reverse current increases drastically and sharply.

- This is the indication that the breakdown has occurred.

- We call this voltage breakdown or Zener voltage, and denoted by V_z .

= How does a Zener diode works in a reverse bias?

- A Zener diode functions similarly to a regular diode when forward biased.

- However, in a reverse-biased mode, a small leakage current flows through the diode.

- As the reverse voltage increases and reaches the predetermined breakdown voltage (V_z),

current begins to flow through the diode.

This current reaches a maximum level determined by the series resistor, after which it stabilizes and remains constant across a wide range of applied voltages.

~ Avalanche breakdown :

- Occurs at a lower doping levels and higher voltages.
- Electrons gain enough energy to create electron-hole pairs.
- Leads to a multiplication effect.

~ Zener breakdown :

- Occurs at high doping levels.
- Electron tunneling through the narrow depletion region.
- Results in a steady reverse current.

→ Applications of Zener diode :

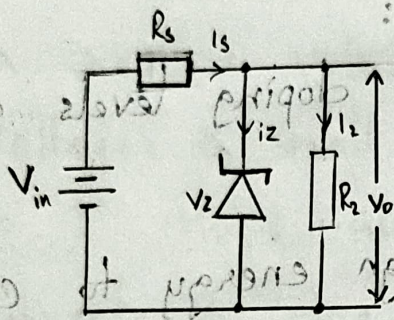
- Zener diode is used as voltage regulator.
- Zener diode is used in over-voltage protections.
- Zener diode is used in clipping circuits.

→ Advantages of Zener diode :

- The size of the Zener diode is so small that it can be used in smaller circuits also in cell phones.
- They are less expensive compared to other diodes.

- It is used for controlling, regulating and stabilizing the voltage in the circuit.
- It has a very high performance standard.

Q. Explain about Zener diode acting as a voltage regulator.



- The ability to keep the reverse voltage across its terminals essentially constant is the key feature of the Zener diode.
- A minimum value of reverse current, I_{ZK} , must be maintained in order to keep the diode in breakdown for voltage regulation.
- When the reverse current is reduced below the knee of the curve, the voltage decreases drastically and regulation is lost.
- A maximum current, I_{ZM} , above which the diode may be damaged due to excessive power dissipation.

— written again after
Exponential diode model

Resistance and Capacitance in P-n Junction diode

* Diode Resistance :

Diode resistance is the effective opposition of a diode which is offered to the flow of current.

• Forward Resistance :-

• Even with forward biasing, a diode won't conduct until it reaches minimum threshold voltage.

Once the applied voltage exceeds this threshold the diode starts to conduct. The resistance offered by the diode in this condition is called forward resistance.

• In forward bias p-n junction diode takes place in two types of resistance based on the voltage applied.

They are :

i) Static resistance or DC resistance

ii) Dynamic resistance or AC resistance

i) Static Resistance :

• The resistance offered by p-n junction diode when it is connected to a DC circuit is called static resistance.

• Static resistance is also defined as the resistance ratio of DC voltage applied across diode to the DC current.

- The resistance offered by the p-n junction diode under forward biased condition is denoted as R_f

$$R_f = \frac{\text{DC Voltage}}{\text{DC Current}}$$

ii) Dynamic Resistance :

• The dynamic resistance is the resistance offered by the p-n junction diode when AC voltages are applied.

- Dynamic resistance is also called as the ratio of change in voltage to the change in current. It is denoted as r_f .

$$r_f = \frac{\text{change in Voltage}}{\text{change in Current}}$$

• Reverse Resistance :

- Reverse resistance is the resistance offered by the p-n junction diode when reverse voltage is applied.
- When reverse biased voltage is applied to the p-n junction diode, the width of depletion layer increases. The depletion region acts as a barrier to the electric current. Hence, a large amount of electric current is blocked by the depletion region.

- Thus, reverse biased diode offer large resistance to the electric current.
- The resistance offered by the reverse biased p-n junction diode is very large compared to the forward biased diode.
- The reverse resistance is in the range of mega ohms ($m\Omega$)

* Diode Capacitance :

A p-n junction diode exhibits two types of capacitance.

- i) Depletion Capacitance under reverse bias and
- ii) Diffusion Capacitance under forward bias.

i) Depletion Capacitance (Junction/ Transition Capacitance)

- Dominant when the diode is reverse biased.
- The depletion region behaves like the dielectric of a capacitor.

$$C_T = \frac{\epsilon A}{w}$$

where A is Junction area, w is depletion width and ϵ is Semi Conductor permittivity.

As reverse voltage increases the depletion width increases the junction capacitance decreases.

ii) Diffusion Capacitance, C_D

Also called as storage capacitance.

- Dominant when the diode is forward biased;
- It results from stored charge due to injected minority carriers.

$$\text{Formulae: } C_D = dQ/dV$$

dQ = change in stored minority carrier charge

dV = change in forward voltage.

- Diffusion Capacitance is much larger than depletion capacitance in forward bias and increases exponentially as forward current or voltage increases.

~ Diode Models:

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A diode model is a simplified representation of the actual diode used to analyse electronic circuits easily.

The commonly used diode models are:

1. Ideal Diode Model
2. Practical Diode Model [Constant voltage drop model]
3. Complete or piecewise linear diode model.
4. Exponential diode model.

① Ideal Diode Model:

In this model:

- Forward biased diode: Acts as a short circuit
- $V_D = 0$
- Reverse biased diode: Acts as an open circuit
- $I_D = 0$
- Advantage: Very simple circuit analysis
- Limitation: Does not represent the practical voltage drop across a diode.

② Practical Diode Model (Constant Voltage Drop model)

A practical diode requires a minimum voltage before significant conduction begins. $\approx 0.7V$

- Forward bias:
- Silicon diode: approx $0.7V$
- Germanium diode: approx $0.3V$

The diode is represented by an ideal diode in series with voltage source.

- Reverse bias:

The diode acts approx as an open circuit.

③ Complete or piecewise linear diode model

This model considers:

- Cut-in Voltage V_s
- Dynamic or forward resistance r_d
- Forward bias:

$$V_D = V_s + I_D r_d$$

for silicon, $V_D = 0.7 + I_D r_d$

This model provides better accuracy than the ideal and practical models.

• Reverse bias:

The diode acts approx. as an open circuit.

④ Exponential Diode Model :

This is most accurate mathematical model based on the diode equation.

$$I_D = I_S (e^{V_D / \eta V_T} - 1)$$

where,

I_D = diode current

I_S = reverse saturation current

V_D = diode voltage

η = ideality factor

V_T = thermal voltage (approx 26mV at room temp)

— Zener diode acting as a voltage regulator.

i) A Zener diode can be used as a voltage regulator to maintain an almost constant output voltage even when the input voltage or load current changes.

ii) A Zener diode is connected in reverse bias across the load while a series resistor R_S is connected between the supply and the load.

Working

iii) When the reverse voltage reaches the Zener breakdown voltage (V_Z). The Zener diode conducts and keeps the voltage across the load approx constant.

$$V_O \approx V_Z$$

The series resistor limits the current:

$$I_S = (V_{in} - V_Z) / R_S$$

and the current divides as:

$$I_S = I_Z + I_L$$

where I_Z is the Zener current
and I_L is the load current.

- iv) If input voltage increases, the extra current flows through the Zener diode, while V_{out} remains nearly V_Z .
 - v) If input voltage decreases, the Zener current decreases, again keeping the output nearly constant.
 - vi) Similarly, changes in load current are compensated by opposite changes in Zener current
-